

✓ Historical Stock Data

```
import pandas as pd
```

```
from google.colab import drive, files
drive.mount('/content/drive')
files.upload()
url_historical_stock = "/content/historical_stock_data_v3.csv"
historical_stock = pd.read_csv(url_historical_stock)
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call `drive.mount('/content/drive')`

Choose Files historical_st...data_v3.csv

historical_stock_data_v3.csv(text/csv) - 11321407 bytes, last modified: 3/2/2026 - 100% done

Saving historical_stock_data_v3.csv to historical_stock_data_v3 (2).csv

✓ Historical Stock data : Ungrouped descriptive statistics

```
(
    historical_stock
    .info()
)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 188212 entries, 0 to 188211
Data columns (total 10 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Symbol      188212 non-null object
1   Year        187844 non-null float64
2   Date        187844 non-null object
3   Open        187844 non-null float64
4   High        187844 non-null float64
5   Low         187844 non-null float64
6   Close       187844 non-null float64
7   Volume      187844 non-null float64
8   Dividend    2440 non-null  float64
9   Split       28 non-null    object
dtypes: float64(7), object(3)
memory usage: 14.4+ MB
```

```
stock_vars = ["Open", "High", "Low", "Close", "Volume", "Dividend"]
```

```
historical_stock[stock_vars].describe()
```

	Open	High	Low	Close	Volume
count	187844.000000	187844.000000	187844.000000	187844.000000	1.878440e+05
mean	170.386701	172.143744	168.550411	170.341924	4.789184e+06
std	420.399416	424.815038	415.982181	420.267384	1.816019e+07
min	1.010000	1.040000	0.970000	0.990000	5.100000e+03
25%	46.750000	47.247500	46.237500	46.727500	8.705000e+05
50%	91.570000	92.590000	90.590000	91.560000	1.774050e+06
75%	178.585000	180.480000	176.650000	178.550000	4.048275e+06
max	9914.170000	9964.770000	9794.000000	9924.400000	1.142269e+09

✓ Historical Stock data : Grouped descriptive statistics

- By Date
- By Symbol

```
(
    historical_stock
    .groupby(["Date"])[["Open", "High", "Low", "Close", "Volume", "Dividend"]]
    .describe()
)
```

Date	Open								High
	count	mean	std	min	25%	50%	75%	max	cour
2024-01-02	604.0	154.892470	357.182667	1.79	43.2375	87.360	158.0000	6901.21	604
2024-01-03	604.0	154.438113	355.560891	1.87	43.9150	86.715	160.6025	6890.00	604
2024-01-04	604.0	153.166854	355.247192	1.80	43.4700	85.720	159.9275	6925.00	604
2024-01-05	604.0	152.866291	354.623554	1.72	43.3000	85.805	157.2925	6900.95	604
2024-01-08	604.0	153.212483	356.431344	1.70	43.6325	85.340	158.6000	6975.00	604
...	...	...	...	...	...	...	...	...	...
2025-03-24	604.0	170.943974	408.955644	1.57	47.2325	89.815	182.7475	7229.46	604
2025-03-25	604.0	172.594785	412.900907	1.56	47.4600	90.440	184.5650	7302.07	604
2025-03-26	604.0	172.526606	414.271412	1.47	47.7425	89.785	184.2375	7305.02	604
2025-03-27	604.0	171.761912	414.008970	1.42	46.9850	90.355	182.5350	7325.95	604
2025-03-28	604.0	171.689603	415.890986	1.46	46.9025	90.045	179.6775	7330.00	604

311 rows × 48 columns

```
(
    historical_stock
    .groupby(["Symbol"])[["Open", "High", "Low", "Close", "Volume", "Dividend"]]
    .describe()
)
```

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✓ Stock Return

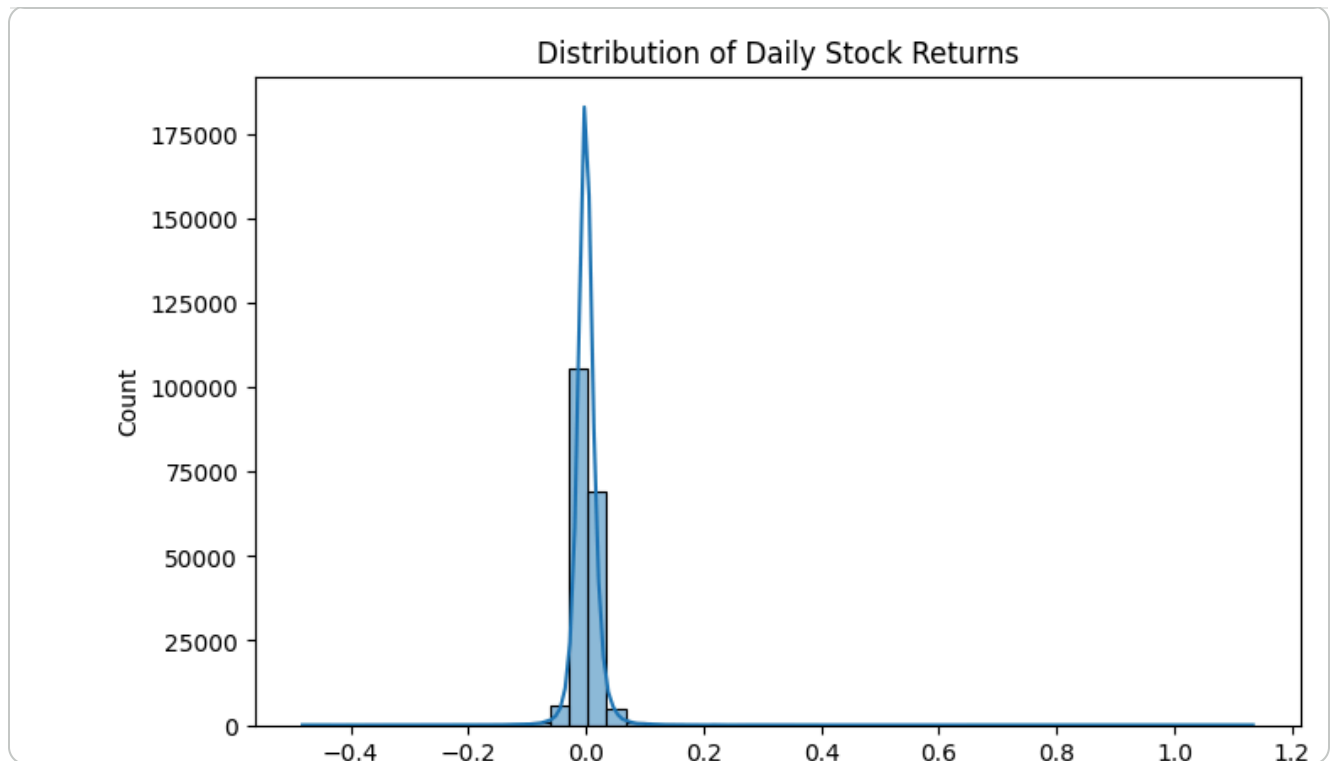
```
historical_stock["daily_return"] = historical_stock.groupby("Symbol")["Close"].
```

```
/tmp/ipykernel_785/2863967872.py:1: FutureWarning: The default fill_method='ffill'
historical_stock["daily_return"] = historical_stock.groupby("Symbol")["Close"]
```

✓ Stock Return : Distribution plot (Histogram)

```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))
sns.histplot(historical_stock["daily_return"].dropna(), bins=50, kde=True)
plt.title("Distribution of Daily Stock Returns")
plt.show()
```



✓ ESG Data 2024

```
url_2024 = "https://bcdanl.github.io/data/esg_proj_2024_data.csv"
esg_proj_2024_data = pd.read_csv(url_2024)
esg_proj_2024_data
```

	Year	Symbol	Name	Sector	Industry	Country	Market_Cap
0	2024	A	Agilent Technologies Inc. Common Stock	Industrials	Biotechnology: Laboratory Analytical Instruments	United States	4036543481
1	2024	AA	Alcoa Corporation Common Stock	Industrials	Aluminum	United States	662213555
2	2024	AAL	American Airlines Group Inc. Common Stock	Consumer Discretionary	Air Freight/Delivery Services	United States	908802460
3	2024	AAP	Advance Auto Parts Inc.	Consumer Discretionary	Auto & Home Supply Stores	United States	447466529
4	2024	AAPL	Apple Inc. Common Stock	Technology	Computer Manufacturing	United States	261431000000
...	...	...	...	...	...	...	...
620	2024	XYL	Xylem Inc. Common Stock New	Industrials	Fluid Controls	United States	3201040268
621	2024	YUM	Yum! Brands Inc.	Consumer Discretionary	Restaurants	United States	3988504441
622	2024	Z	Zillow Group Inc. Class C Capital Stock	Consumer Discretionary	Business Services	United States	1019546912
623	2024	ZBH	Zimmer Biomet Holdings Inc. Common Stock	Health Care	Industrial Specialties	United States	2447677802
624	2024	ZTS	Zoetis Inc. Class A Common Stock	Health Care	Biotechnology: Pharmaceutical Preparations	United States	7253530835
625 rows × 13 columns							
Next steps: New interactive sheet							

✓ ESG 2024 : Ungrouped descriptive statistics

```
esg_vars = ["Total_ESG", "Environmental", "Social", "Governance", "Controversy"]  
  
esg_proj_2024_data[esg_vars].describe()
```

	Total_ESG	Environmental	Social	Governance	Controversy	
count	625.000000	602.000000	602.000000	602.000000	573.000000	
mean	21.567040	5.747342	9.017774	6.819103	1.942408	
std	7.041637	5.200518	3.567490	2.371114	0.801113	
min	6.400000	0.000000	0.800000	2.400000	0.000000	
25%	16.300000	1.800000	6.700000	5.225000	1.000000	
50%	21.100000	3.950000	8.900000	6.300000	2.000000	
75%	26.000000	8.875000	11.175000	7.900000	2.000000	
max	52.000000	25.300000	22.500000	19.400000	5.000000	

✓ ESG 2024 : Grouped descriptive statistics

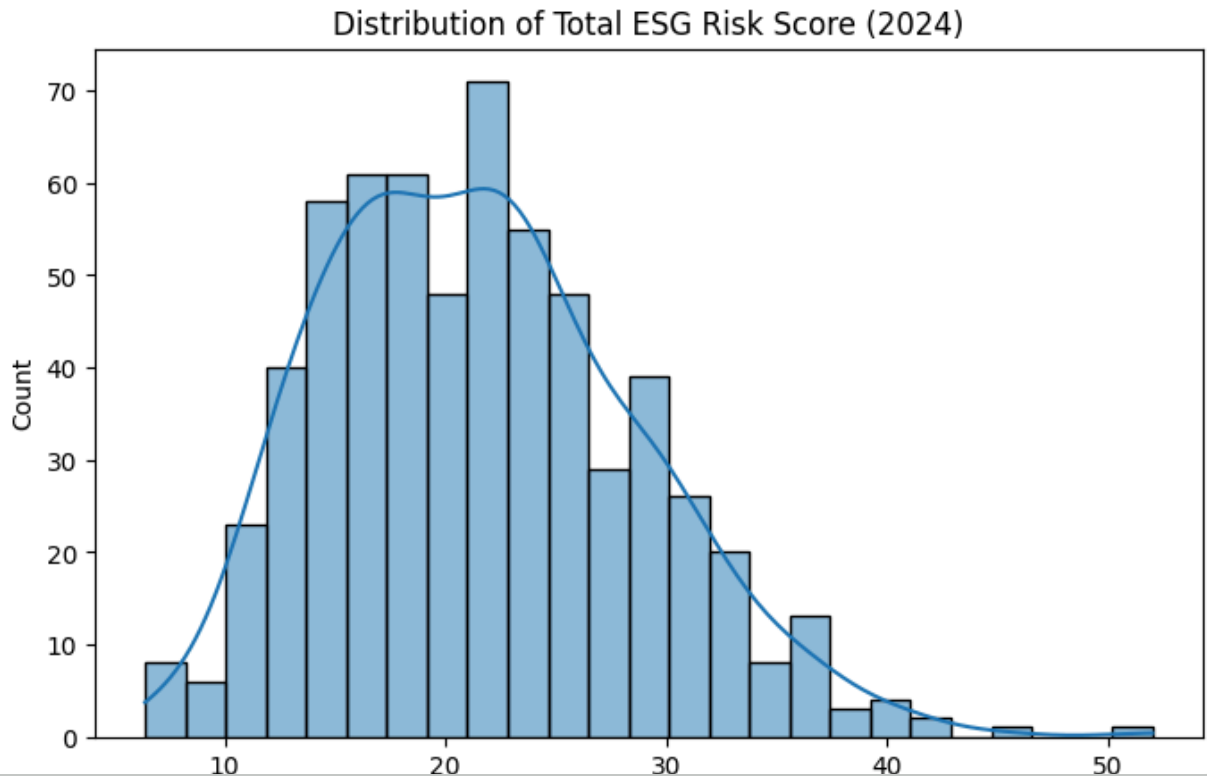
By Sector and Industry

```
(  
    esg_proj_2024_data  
    .groupby(["Sector","Industry"])[["Total_ESG", "Environmental", "Social", "C  
    .describe()  
)
```

		Total_ESG							
		count	mean	std	min	25%	50%	75%	
Sector	Industry								
Basic Materials	Forest Products	1.0	24.200000	NaN	24.2	24.200	24.20	24.2	
	Metal Mining	2.0	29.550000	2.899138	27.5	28.525	29.55	30.5	
	Precious Metals	5.0	18.140000	10.750953	7.3	8.200	20.50	21.4	
Consumer discretionary	Advertising	1.0	16.300000	NaN	16.3	16.300	16.30	16.3	
	Air Freight/Delivery Services	5.0	27.140000	4.491436	19.7	26.400	28.60	30.3	
...	...	...	...	...	...	...	...	...	
Utilities	Natural Gas Distribution	7.0	29.228571	11.278108	19.0	22.650	23.60	32.3	
	Oil & Gas Production	1.0	25.100000	NaN	25.1	25.100	25.10	25.1	
	Oil/Gas Transmission	2.0	28.700000	8.343860	22.8	25.750	28.70	31.6	
	Power Generation	11.0	25.600000	2.794280	21.4	23.350	26.50	27.2	
	Water Supply	2.0	23.100000	2.121320	21.6	22.350	23.10	23.8	
rows × 40 columns									

✓ ESG 2024 : Distribution plot (Histogram)

```
plt.figure(figsize=(8,5))
sns.histplot(esg_proj_2024_data["Total_ESG"], kde=True, bins=25)
plt.title("Distribution of Total ESG Risk Score (2024)")
plt.show()
```

✓ ESG Data 2025

```
from google.colab import drive, files
drive.mount('/content/drive')
files.upload()
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call `drive.mount('/content/drive', force_remount=True)`

`esg_proj_2025.csv`

esg_proj_2025.csv(text/csv) - 62838 bytes, last modified: 3/1/2026 - 100% done

Saving `esg_proj_2025.csv` to `esg_proj_2025 (1).csv`

`{'esg_proj_2025 (1).csv':`
`b'Symbol,total_esg,environmental,social,governance,controversy\r\nA,10.1,1.1,5.0`

```
url_2025 = "/content/esg_proj_2025.csv"
esg_proj_2025 = pd.read_csv(url_2025)
esg_proj_2025
```

	Symbol	total_esg	environmental	social	governance	controversy	
0	A	10.1	1.1	5.0	3.9	1.0	
1	AA	25.1	14.7	8.0	2.4	3.0	
2	AADI	26.4	NaN	NaN	NaN	NaN	
3	AAL	23.7	9.6	11.3	2.8	2.0	
4	AAMI	31.3	NaN	NaN	NaN	NaN	
...	...	...	...	...	...	...	
3142	FLYW	19.6	NaN	NaN	NaN	NaN	
3143	BIDU	18.0	2.9	8.1	7.0	3.0	
3144	SMBK	30.6	NaN	NaN	NaN	NaN	
3145	ESNT	22.3	NaN	NaN	NaN	NaN	
3146	KOP	24.1	NaN	NaN	NaN	1.0	

3147 rows × 6 columns

Next steps:

New interactive sheet

```
esg_proj_2025["Year"] = 2025
esg_proj_2025
```

	Symbol	total_esg	environmental	social	governance	controversy	year	Y
0	A	10.1	1.1	5.0	3.9	1.0	2025	2
1	AA	25.1	14.7	8.0	2.4	3.0	2025	2
2	AADI	26.4	NaN	NaN	NaN	NaN	2025	2
3	AAL	23.7	9.6	11.3	2.8	2.0	2025	2
4	AAMI	31.3	NaN	NaN	NaN	NaN	2025	2
...	...	...	...	...	...	...	...	...
3142	FLYW	19.6	NaN	NaN	NaN	NaN	2025	2
3143	BIDU	18.0	2.9	8.1	7.0	3.0	2025	2
3144	SMBK	30.6	NaN	NaN	NaN	NaN	2025	2
3145	ESNT	22.3	NaN	NaN	NaN	NaN	2025	2
3146	KOP	24.1	NaN	NaN	NaN	1.0	2025	2

3147 rows × 8 columns

Next steps: [New interactive sheet](#)

```
(
    esg_proj_2025
    .info()
)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3147 entries, 0 to 3146
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Symbol          3147 non-null   object
1   total_esg       3147 non-null   float64
2   environmental    1158 non-null   float64
3   social          1158 non-null   float64
4   governance       1158 non-null   float64
5   controversy     1761 non-null   float64
6   year            3147 non-null   int64
7   Year            3147 non-null   int64
dtypes: float64(5), int64(2), object(1)
memory usage: 196.8+ KB
```

✓ ESG 2025 : Ungrouped descriptive statistics

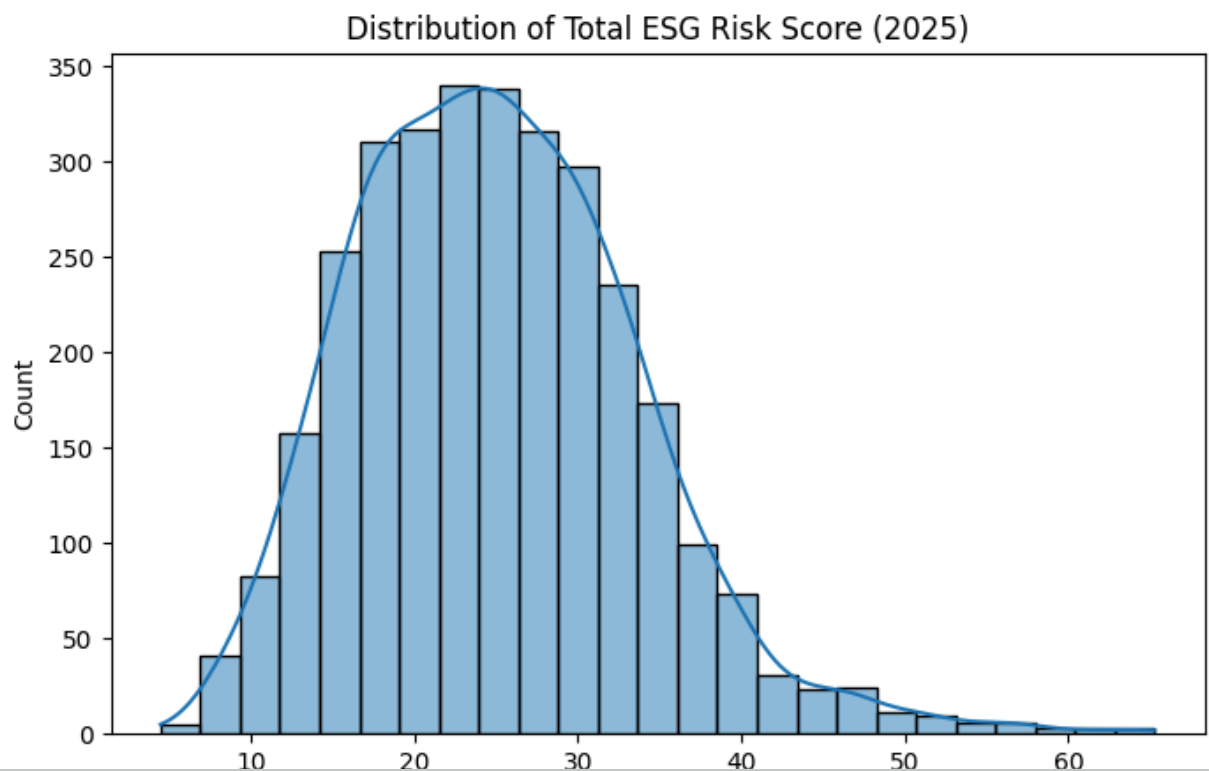
```
esg_vars2 = ["total_esg", "environmental", "social", "governance", "controversy"]

esg_proj_2025[esg_vars2].describe()
```

	total_esg	environmental	social	governance	controversy
count	3147.000000	1158.000000	1158.000000	1158.000000	1761.000000
mean	25.005084	6.166149	9.505613	5.771071	1.582056
std	8.529579	5.511283	3.807264	2.529303	0.696743
min	4.500000	0.000000	0.300000	0.900000	1.000000
25%	18.600000	1.800000	7.100000	4.100000	1.000000
50%	24.500000	4.400000	9.400000	5.200000	1.000000
75%	30.400000	9.400000	11.800000	6.900000	2.000000
max	65.300000	32.500000	27.900000	18.600000	5.000000

✓ ESG 2025 : Distribution plot (Histogram)

```
plt.figure(figsize=(8,5))
sns.histplot(esg_proj_2025["total_esg"], kde=True, bins=25)
plt.title("Distribution of Total ESG Risk Score (2025)")
plt.show()
```



✓ ESG 2024 + 2025

```
common_symbols = set(esg_proj_2024_data["Symbol"]).intersection(
    set(esg_proj_2025["Symbol"]))
len(common_symbols)
```

625

```
esg_proj_2025 = esg_proj_2025.rename(columns={ "total_esg": "Total_ESG", "envir
```

```
esg_2024_common = esg_proj_2024_data[
    esg_proj_2024_data["Symbol"].isin(common_symbols)
]
```

```
esg_2025_common = esg_proj_2025[
    esg_proj_2025["Symbol"].isin(common_symbols)
]
```

```
esg_2024_common = esg_2024_common.drop_duplicates()
esg_2025_common = esg_2025_common.drop_duplicates()
```

```
esg_2024_2025 = pd.concat(
    [esg_2024_common, esg_2025_common],
    ignore_index=True
)
esg_2024_2025 = esg_2024_2025.drop(columns=["year"])

esg_2024_2025
```

1	Name	Sector	Industry	Country	Market_Cap
A	Agilent Technologies Inc. Common Stock	Industrials	Biotechnology: Laboratory Analytical Instruments	United States	4.036543e+10
A	Alcoa Corporation Common Stock	Industrials	Aluminum	United States	6.622136e+09
L	American Airlines Group Inc. Common Stock	Consumer Discretionary	Air Freight/Delivery Services	United States	9.088025e+09
D	Advance Auto Parts Inc.	Consumer Discretionary	Auto & Home Supply Stores	United States	4.474665e+09
L	Apple Inc. Common Stock	Technology	Computer Manufacturing	United States	2.614310e+12
...	...	...	...	...	...
3	NaN	NaN	NaN	NaN	NaN
D	NaN	NaN	NaN	NaN	NaN
A	NaN	NaN	NaN	NaN	NaN
D	NaN	NaN	NaN	NaN	NaN
J	NaN	NaN	NaN	NaN	NaN
ns					

Next steps:

New interactive sheet

✓ Exploratory Data Analysis

Did the average ESG score change from 2024 to 2025?

```
esg_2024_2025.groupby("Year")["Total_ESG"].describe()
```

	count	mean	std	min	25%	50%	75%	max	
Year									
2024	625.0	21.56704	7.041637	6.4	16.3	21.1	26.0	52.0	

```
plt.figure(figsize=(6,5))
sns.boxplot(data=esg_2024_2025, x="Year", y="Total_ESG")
plt.title("Total ESG Risk Score Distribution by Year")
plt.show()
```

Total ESG Risk Score Distribution by Year